



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CCS339 Cryptocurrency and Blockchain technologies

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit II / FORTH – the precursor for Bitcoin scripting

Course Outcome: CO 2

Unit Learning Outcome: UO 2b

Activity Chosen: Learning by Teaching

Justification:

- The primary goal of the learning-by-teaching activity was to foster active learning and consolidate students' knowledge by having them teach assigned topics to their peers. This approach aimed to enhance their understanding of the material, as explaining concepts requires a deep comprehension and thorough review. Additionally, it sought to improve students' communication skills by having them articulate their knowledge clearly and effectively. The activity promoted active participation, transforming students from passive recipients to active contributors in the learning process, and fostered a collaborative environment where peers could learn from each other.

Time Allotted for the Activity: 35 minutes

Details of the Implementation:

- Prior to the session, the instructor asked the students about their willingness to teach a particular topic.
- Devishenbhaga R expressed interest in taking the class.
- The student was assigned the topic "FORTH – the precursor for Bitcoin scripting" based on its relevance and complexity.
- The student was given time to prepare the assigned topic and was encouraged to gather information from various sources, organize the content effectively, and plan her teaching approach.
- During the session, students took turns presenting their assigned topics to their classmates.
- After the presentations, a question-and-answer session was held to encourage active participation and clarify any doubts.
- This allowed students to engage in discussions and exchange additional insights related to the topic.
- Finally, the instructor summarized the content.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PSO1
CO 2	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	1	1	1	1	1
Justification for correlation	Students understood the fundamental concept of Bitcoin scripting by applying the knowledge of mathematics.	Students analyzed blockchain use cases and scenarios using engineering sciences.	Students gained proficiency in writing Bitcoin scripting	Students grasped the concept by investigate complex problems	Students will be able to apply their knowledge of Bitcoin Scripts to excel in software development.

• **Images / Screenshot of the practice:**



Figure 1: Discussion about FORTH – the precursor for Bitcoin scripting by Devishenbhaga R

Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- Presenting the seminar helped to understand the topic more deeply as it had to explain it to others.
- Students found the seminar activity to be a valuable learning experience that enhanced their understanding, communication skills, and confidence

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- By involving students in teaching and discussions, the seminar promotes active learning, making the educational experience more dynamic and interactive.

- Students spent their time in self-learning.
- The seminar provides an opportunity for students to receive feedback from both peers and instructors, helping them identify strengths and areas for improvement.
- ❖ *Challenges faced in implementation:*
 - Due to shyness, some students hesitate to teach.
 - Keeping all students engaged throughout the seminar can be challenging, especially if presentations are lengthy or not interactive.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching/>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5_VA_CS3362_Learning%20by%20teaching.pdf
4. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/Unit_5_Learning_by_Teaching.pdf

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